

# GABRYEL VERÍSSIMO

Aspiring Backend Software Engineer · Computer Science Student

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## PROFILE

Computer Science student (Year 3, University of Greenwich) building production-grade backend systems in Python and FastAPI, with a focus on fintech APIs — payments, ledgers, idempotent transfers, and secure authentication. Comfortable with PostgreSQL, REST design, Docker, and test-driven development. Seeking a junior or placement backend developer role in London.

## TECHNICAL SKILLS

**Languages:** Python, C#, SQL, JavaScript, C++, HTML/CSS

**Backend:** FastAPI, REST APIs, PostgreSQL, Redis, SQLAlchemy, Alembic, JWT authentication

**Tools & Practices:** Docker, Git, GitHub Actions (CI), pytest, test-driven development, background workers & queues, Linux

## PROJECTS

### PayLedger — Double-Entry Payments API

*Python · FastAPI · PostgreSQL · Redis · Docker | [github.com/gabryelvs/payledger](https://github.com/gabryelvs/payledger) | [live demo](#)*

- Backend payment platform built around an immutable double-entry ledger: balances and transfers recorded as balanced, append-only ledger entries.
- Race-safe transfers via row-level database locking (no lost updates under concurrent load); idempotent writes; money stored as integer minor units to avoid floating-point errors.
- Test-driven (31 tests incl. a concurrency proof), with Docker Compose, GitHub Actions CI, and Alembic migrations.

### FX-Service — Async Currency Exchange API

*Python · FastAPI · httpx · Redis | [github.com/gabryelvs/fx-service](https://github.com/gabryelvs/fx-service) | [live demo](#)*

- Async microservice serving live currency rates and conversions; caches ECB rates in Redis with an in-process scheduled refresh loop.
- Resilient: keeps serving last-known rates when the upstream provider is down (stale fallback), proven by an automated outage test.
- Test-driven (31 tests, 90% coverage) with Docker Compose and GitHub Actions CI.

### Webhook-Dispatcher — Reliable Webhook Delivery

*Python · FastAPI · Redis | [github.com/gabryelvs/webhook-dispatcher](https://github.com/gabryelvs/webhook-dispatcher)*

- Delivers webhooks via a Redis queue and a separate worker process, signing each request (HMAC-SHA256) so receivers can verify authenticity.
- Retries with exponential backoff and jitter, dead-letters exhausted deliveries, and supports manual replay; at-least-once delivery with de-duplication headers.
- Test-driven (28 tests, 92% coverage) with Docker Compose and GitHub Actions CI.

*Academic & coursework projects*

- **Secure Web App with Multi-Factor Authentication** (Python) — built and presented a multi-step secure login system with a classmate.
- **Database Design, Security & Optimisation** (SQL) — designed a database from scratch and hardened an existing one by fixing vulnerabilities.
- Other team projects: led a team of four building an e-commerce site (HTML/CSS/JS); a Python + database family-tree system; and an educational mobile game.

## WORK EXPERIENCE

### Web Designer & Media, Impact Brixton — 2021

- Worked within a team to build a new company website (layout, visual content, UX) and supported social media and digital marketing.

## EDUCATION

**BSc Computer Science** — University of Greenwich (2023–Present): Foundation Year, Year 1, Year 2, currently Year 3.

City of Westminster College (2022–23): BTEC Level 2 ICT · GCSE English. Lambeth College (2021–22): L1 Diploma ICT · GCSE Maths · FS Maths & English. SESI-SENAI College (2019–20): Diploma in Microsoft Applications.

## **LANGUAGES**

**Portuguese:** Native   **English:** Fluent (advanced — speaking, reading, writing)